

FORECASTING DAILY EQUITY VOLATILITY: CAN MACHINE LEARNING BEAT CLASSICAL MODELS?

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ABSTRACT

This project asks a simple question: can machine-learning and deep-learning models forecast next-day stock-market volatility better than classical statistical models? Eight models were tested on 18.5 years of S&P 500 daily returns (2008–2026) using a strict walk-forward setup with no look-ahead bias. EGARCH(1,1), a classical model with a leverage term, gave the best result on the main metric (QLIKE), and the leverage models significantly beat plain GARCH(1,1) on a test window containing the 2022 bear market and the April-2025 tariff shock. The LSTM came third, ahead of plain GARCH and statistically tied with the winner, but no learned model beat the leverage GARCH family where it matters. The well-known result that GARCH-type models are very hard to beat on daily data survives, in leverage-aware form.

Index Terms: Volatility forecasting, GARCH, QLIKE, XGBoost, LSTM

1. INTRODUCTION

Volatility is the size of an asset’s price swings. It matters for option pricing, risk limits and position sizing. The direction of tomorrow’s return is essentially unpredictable, but its size is not: calm days follow calm days and wild days follow wild days. Classical models like GARCH have used this persistence for decades. This project checks whether modern machine-learning (XGBoost) and deep-learning (LSTM) models can do better, testing every model fairly inside one evaluation harness.

2. PROBLEM AND OBJECTIVES

The task is to forecast tomorrow’s variance using only information available today. True volatility is never observed, so forecasts are scored against the squared return r_t^2 , a noisy but unbiased proxy. The objectives were: (i) build honest classical baselines (naive, rolling, EWMA, and the GARCH family); (ii) build a leak-free walk-forward evaluation with RMSE, MAE, QLIKE and the Diebold–Mariano test; (iii) build a walk-forward XGBoost and a regularized LSTM; and (iv) compare all models and test whether any difference is statistically real.

3. METHOD

Baselines. Naive ($\hat{\sigma}_t^2 = r_{t-1}^2$), 21-day rolling variance, and EWMA (RiskMetrics, $\lambda = 0.94$).

GARCH family. GARCH(1,1) models variance as $\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2$. GJR-GARCH and EGARCH add a leverage term so that falls raise volatility more than

Table 1. Leaderboard over the test window, sorted by QLIKE (lower is better).

Model	RMSE	MAE	QLIKE
EGARCH(1,1)	0.00728	0.00556	1.5042
GJR-GARCH(1,1)	0.00731	0.00554	1.5091
LSTM	0.00686	0.00499	1.5287
GARCH(1,1)	0.00751	0.00576	1.5569
EWMA(0.94)	0.00750	0.00573	1.5814
XGBoost	0.00691	0.00496	1.5911
Rolling(21)	0.00762	0.00571	1.6344
Naive	0.00951	0.00672	5440.50

risers. Parameters are re-estimated about monthly and rolled forward daily.

Learned models. XGBoost and an LSTM use the same features (recent returns, squared returns, and 5- and 21-day realized variance) and the same target (log of a 5-day-forward smoothed variance). XGBoost is re-trained about monthly; the LSTM uses train-only scaling and early stopping. Both respect the walk-forward rule: no future data leaks into any forecast.

Scoring. QLIKE is the headline metric because its model ranking is robust to noise in the proxy and it punishes under-prediction of variance. The Diebold–Mariano test checks whether a gap in loss is real or luck.

4. EXPERIMENTS AND RESULTS

Data. S&P 500 daily closes (2008–2026, Yahoo Finance), 4,650 returns. The last 30% (1,395 days, 2020-12-07 to 2026-06-29, including the 2022 bear market and the April-2025 tariff shock) is the out-of-sample test window shared by every model.

Table 1 shows the leaderboard. EGARCH(1,1) wins on QLIKE, statistically tied with GJR-GARCH (Diebold–Mariano $p = 0.47$), and both leverage models significantly beat plain GARCH(1,1) ($p < 0.0001$). The LSTM ranks third, ahead of plain GARCH, and is statistically tied with the winner ($p = 0.19$). The learned models take the point-accuracy metrics (LSTM the best RMSE, XGBoost the best MAE) but rank lower on QLIKE, because their smoothed target makes them accurate on average yet slow to react at volatility spikes. Fig. 1 shows the winning forecast tracking realized volatility.

5. DISCUSSION

The leverage models lead because the test window is dominated by two sharp crashes (2022 and April 2025), and their extra term turns a large fall directly into higher next-day vari-

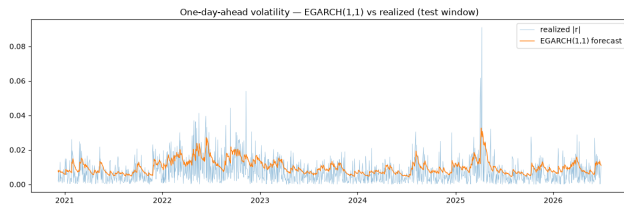


Fig. 1. Best model by QLIKE, EGARCH(1,1), against realized volatility over the test window.

ance; on this window that advantage over plain GARCH is highly significant. The GARCH family as a whole stays ahead of the learned models for three reasons: it is fit by maximum likelihood on the right objective, while the learned models train on a smoother substitute target; daily data is small data, which favours simple models over neural networks; and QLIKE rewards fast reactions to spikes, which the GARCH recursion delivers the very next day. The case against the learned models is narrower than on shorter samples, though: the LSTM learned the leverage pattern from data and finished statistically tied with the winner. The main limitations are the single asset and test window, the noisy squared-return proxy, and modest hyperparameter tuning.

6. CONCLUSION

Under a leak-free walk-forward test on 18.5 years of S&P 500 data, a monthly re-estimated EGARCH(1,1) gave the best one-day-ahead volatility forecasts, the leverage models significantly beat plain GARCH, and no learned model beat the leverage GARCH family on QLIKE. The learned models were competitive (the LSTM statistically tied with the winner) but not better where it counts. The clearest lesson was methodological: a leak-free evaluation and the choice of metric matter more than model complexity. Useful next steps are intraday realized-variance targets, extra inputs such as the VIX, and multi-asset training.

7. ARTIFACTS

Code: <https://github.com/iitgF/T9> (the volforecast package and notebooks). LaTeX source (Overleaf): <https://www.overleaf.com/project/6a48e18e7dee3049b3ee4209>. Full report: `report.tex`. AI-assisted tooling was used for refactoring and drafting; all decisions, checks and the final text are the author's, and every number was reproduced from the committed code.

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